Design, Dynamics, Drive of BLDC Motor

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Why BLDC ?

High energy efficiency High Power to Weight ratio Quick dynamic respone

Design

Basic Terminology

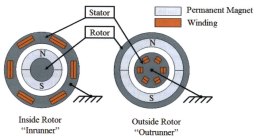

Figure 1: Stator & Rotor, Source:[1]

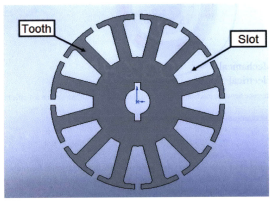

Figure 2: Tooth & Slot, Source:[1]

Air Gap

Slot

Winding Configuration

They have three phase windings and can be connected to each other in wye or delta configuration. Wye has higher torque (theoretically torque constant is greater by a factor $\sqrt{3}$ [2, p. 25]) because in the wye configuration, at any time one of phase is open and other two-phase are in series thus equal current flows through them whereas in delta it gets divided into two phases. Thus more current in each phase causes more torque, one will push and another phase will pull in the wye configuration.

Dynamics

Drive

Simon has done the gr8 job. what he does is actually amazing. [2] This is proper documentation

His work was based on condition.[1]

$$\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla)\vec{u} = -\frac{1}{\rho}\nabla p + \nabla \sigma + \vec{f}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0$$

Testing whether it works..!

References

[1] S. W. Colton, "Design and prototyping methods for brushless motors and motor control," Masters Thesis, Massachusetts Institute of Technology, 2010 [Online]. Available: http://dspace.mit.edu/handle/1721.1/61599

[2] S. Kalouche, "Design for 3d agility and virtual compliance using proprioceptive force control in dynamic legged robots," Masters Thesis, Carnegie Mellon University, 2016.